

HW # 27

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CS-F

(a)
i) ~~4th~~ ^{All} rows A 3rd col B

ii) 3rd row A All cols B

iii) ~~3rd~~ 3rd row A 4th col B

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

(b)
 $\left. \begin{array}{l} \det(A) \neq 0 \\ \det(B) \neq 0 \end{array} \right\} \text{Invertible}$

$$\det(A+B) = \det(A) + \det(B)$$

$\det(A+B) \neq 0$
must not be

unless the det's
are same in magnitude
but opposite in signs

Therefore its not necessary

(c)

$$\begin{bmatrix} 3s & -2 \\ -6 & s \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$3s^2 - 12 = \det(A)$$

$$Ax_1 = \begin{bmatrix} 4 & -2 \\ 1 & s \end{bmatrix}, \det(Ax_1) = 4s + 2$$

$$Ax_2 = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}, \det(Ax_2) = 3s + 24$$

$$x_1 = \frac{\det(Ax_1)}{\det(A)} = \frac{4s+2}{3s^2-12}$$

$$x_2 = \frac{\det(Ax_2)}{\det(A)} = \frac{3s+24}{3s^2-12} \quad s \neq \pm 2 \rightarrow \text{solution is unique}$$

Solution is unique if $x_1 = x_2 = 0$

$$4s+2=0$$

$$4s = -2$$

$$s = -\frac{1}{2}$$

$$3s+24=0$$

$$s = -\frac{24}{3}$$

$$s = -8$$

$$x_1 =$$

(e)

$$E_1 E_2 E_3 A = I$$

$$E_1 E_2 E_3 I^{-1} = A^{-1}$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A^{-1}$$

$$\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A^{-1}$$

$$\begin{bmatrix} -1 & 5 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A^{-1}$$

$$\begin{bmatrix} -1 & 5 \\ 1 & 0 \end{bmatrix} = A^{-1}$$

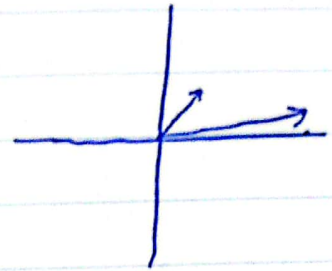
(d)

Area of parallelogram :

$$\det \begin{bmatrix} b_1 & b_2 \end{bmatrix} = \begin{vmatrix} 1 & 5 \\ 3 & 1 \end{vmatrix}$$

$$= 1 - 15$$

$$= -14$$



$$\text{Imge} = |\text{Area}| * \det(A)$$

$$\text{of S Area} = 14 * (0)$$

$$= 0$$

This shows that the transformation made it a line or a point instead of a parallelogram (two diff lines)